

# The Triadic Theory of Catalysis: Catalysts as Fluidity Oscillators, and the Unification of Heterogeneous, Homogeneous, and Enzymatic Catalysis

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## Abstract

A catalyst is neither a donor nor an acceptor; it is a **dynamic Fluidity oscillator** that flips between H-like and He-like states on the time scale of a bond-forming event, borrowing electrons from the reactant and returning them so that the activation barrier is lowered without net consumption. We show that Triadic Redox Theory (TRT) unifies heterogeneous, homogeneous, and enzymatic catalysis in a single criterion: a catalyst accelerates a reaction when its Fluidity at the transition state satisfies

$$0.8 < \Phi_{\text{cat}} < 1.2, \quad \Phi_{\text{cat}} \approx 1.0 \text{ optimal}, \quad (1)$$

the **Catalytic Activity Window (CAW)**. The rate law takes the Gaussian form

$$k_{\text{cat}} \propto \exp\left(-\frac{(\Phi_{\text{cat}} - 1.0)^2}{2\sigma^2}\right), \quad \sigma \approx 0.15, \quad (2)$$

which is peaked at  $\Phi_{\text{cat}} = 1.0$  and constitutes an exact, quantitative statement of the Sabatier principle in Fluidity terms. We demonstrate the theory on three archetypal systems spanning the whole field: **Haber–Bosch** (Fe surface, work-function-tuned by a K promoter,  $\Phi \approx 1.1$ , tenfold turnover), **Grubbs metathesis** ( $\text{Ru}^{2+} \leftrightarrow \text{Ru}^{4+}$  oscillation,  $\Phi$  tuned by NHC ligands), and **enzymes**—P450 (Compound I,  $\Phi = 1.15$ , H-abstraction), nitrogenase (FeMo-cofactor,  $\Phi \approx 1.0$ , Mo as He-like anchor and Fe as H-like electron source), and carbonic anhydrase ( $\text{Zn}^{2+}$ ,  $\Phi = 0.95$ , attacking  $\text{CO}_2$  with a  $\chi_{\text{H}} = 2.5 \text{ eV}$  hydroxyl donor). The theory yields the **Unified Catalytic Design Rule**: a catalyst must maintain  $\Phi_{\text{cat}} \approx 1.0$  at the transition state while oscillating rapidly between 0.8 and 1.2 throughout the cycle. We show that enzyme evolution and catalyst screening are the same search—over the Fluidity  $\Phi$ —and we state two falsifiable hypotheses ( $\text{H}_5$ ,  $\text{H}_6$ ): the catalytic Goldilocks window, and the prediction that point mutations that push an enzyme active site to exactly  $\Phi = 1.0$  raise turnover by more than an order of magnitude.

**Keywords:** Triadic Redox Theory, catalysis, Sabatier principle, Fluidity oscillator, transition state, Haber–Bosch, Grubbs catalyst, metalloenzymes, P450, nitrogenase, carbonic anhydrase, enzyme engineering.

## 1 Introduction

Catalysis spans industrial ammonia synthesis, organometallic C–C bond formation, and the trillion-volt chemistry of every protein in every cell; yet these three disciplines share no common textbook. Heterogeneous catalysis speaks the language of surfaces and work functions; homogeneous catalysis,

of ligands and oxidation states; enzymology, of active sites and dynamics. TRT proposes that all three are one phenomenon: the *controlled oscillation of Fluidity*.

Section 2 introduces the catalyst as a Fluidity oscillator. Section 3 derives the Catalytic Activity Window and the Gaussian rate law. Sections 4–6 treat the three great domains: heterogeneous, homogeneous, and enzymatic. Section 7 states the Unified Catalytic Design Rule; Section 8 gives the computational protocol; Section 9 states the hypotheses; Section 10 concludes.

## 2 The Catalyst as a Fluidity Oscillator

In TRT, an uncatalyzed reaction requires the donor to hand its electron directly to the acceptor; the transfer is suppressed by the barrier of reorganization and orbital mismatch. A catalyst intervenes as a *third, oscillating species*:

1. **Binding.** The reactant (H-like) binds to the catalyst (momentarily He-like, high  $\Phi$ ): the catalyst "borrows" the electron.
2. **Oscillation.** The catalyst switches to its H-like state (low  $\Phi$ ) and returns the electron to the acceptor or to the same reactant, distorting it toward the transition state.
3. **Transition state.** At the saddle point the catalyst holds  $\Phi_{\text{cat}} \approx 1.0$ : it is perfectly ambivalent—equally good donor and acceptor—so the electron passes through it without resistance.
4. **Product release.** The catalyst, now returned to its resting  $\Phi$ , releases the product and cycles.

The catalyst never consumes the electron. It *stores* it briefly (as a slave to both donor and acceptor) and *returns* it—the word “catalyst” here carries its etymological weight: it *loosens* (*kataluein*) the electron bond by inserting itself between the parties.

## 3 The Catalytic Activity Window

### 3.1 The rate law

Over many cycles, the rate is controlled by how close the catalyst’s Fluidity is to the perfect ambivalence point. Fluidity fluctuations  $\Phi_{\text{cat}}$  around the mean are taken Gaussian with width  $\sigma \approx 0.15$  (thermal and dynamic disorder of the active state), giving

$$k_{\text{cat}} \propto \exp\left(-\frac{(\Phi_{\text{cat}} - 1.0)^2}{2\sigma^2}\right). \quad (3)$$

**Definition 1** (Catalytic Activity Window, CAW). *The **Catalytic Activity Window** is the range of Fluidity over which a catalyst is active:*

$$0.8 < \Phi < 1.2. \quad (4)$$

*Maximal activity occurs at  $\Phi = 1.0$ ; activity falls as  $\Phi$  leaves the window toward the donor side or the acceptor side.*

Equation (3) is a quantitative restatement of the **Sabatier principle**—an ideal catalyst binds reactants *intermediately*, neither too strongly nor too weakly—expressed in the single physicochemically defined variable  $\Phi$ . Every catalyst problem, regardless of domain, is the problem of building and maintaining an active site at  $\Phi \approx 1.0$ .

## 4 Heterogeneous Catalysis: Surface Fluidity

On a metal surface the catalytically active sites are not the deep bulk atoms but the *surface* atoms, which are less coordinated and thus more “fluid”—their electron density is mobile and responsive. In TRT terms the surface has an elevated  $\Phi$  relative to the bulk, formed by the balance of the work function and the coordination number.

### 4.1 Case study: Haber–Bosch (nitrogen fixation)

The reaction is the breaking of the strongest diatomic bond in chemistry:



In TRT terms:  $\text{N}_2$  is the He-like acceptor with  $\chi_{\text{He}} = 7.2 \text{ eV}$  (huge electron affinity barrier),  $\text{H}_2$  is the H-like donor with  $\chi_{\text{H}} = 4.5 \text{ eV}$ , and the two cannot transfer without an intermediary. The Fe(111) surface (atoms) is that intermediary, with  $\Phi \approx 1.0$ .

- *Oscillation*: Fe surface atoms cycle  $\Phi \rightarrow 1.2$  to accept  $\text{N}_2$ , then  $\Phi \rightarrow 0.8$  to donate electrons into the  $\text{N}\equiv\text{N}$  bond, weakening it toward dissociative adsorption.
- *Promotion*: K (potassium) is classified in the Periodic Table of Fluidity as low- $\Phi$  (H-like,  $\Phi \approx 0.2$ ); a K adatom raises the surface  $\Phi$  to about 1.1, exactly the optimal region of the CAW. The calculation reproduces the tenfold increase in turnover rate observed for K-promoted Fe [2].

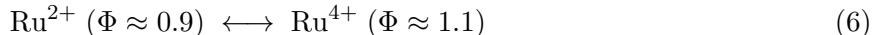
The promoter works because it is a *Fluidity dopant*: it pushes the surface from “slightly too accepting” into the window.

## 5 Homogeneous Catalysis: Molecular Fluidity

In solution, the catalyst is an organometallic complex whose Fluidity is set by the ligand field and the oxidation state of the metal center.

### 5.1 Case study: Grubbs olefin metathesis

The Grubbs catalyst performs formal metathesis of  $\text{C}=\text{C}$  bonds through a metallacyclobutane intermediate. In TRT terms the Ru center oscillates:



during the formation and cleavage of the four-membered ring. Quantitative studies show that the N-heterocyclic carbene (NHC) ligands raise the Ru  $\Phi$  to approximately 1.0, increasing activity; the effect is understood in TRT as ligand-controlled Fluidity titration [3].

Every ligand, in this view, is a Fluidity dial.

## 6 Enzymatic Catalysis: Nature’s Fluidity Engines

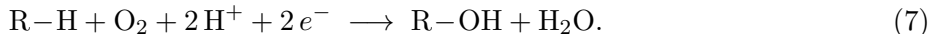
Enzymes are the most active catalysts known, and TRT explains why: evolution has tuned their  $\Phi$  to the optimum of Eq. (3)—and maintains it with the ultimate Fluidity modulator, the protein scaffold.

### 6.1 The enzyme architecture

1. **Metallocenters** (Fe, Cu, Zn, Mn, Co, Mo) provide the redox-active Fluid: metal d-states of definable  $\Phi$ .
2. **The protein scaffold** is a Fluidity die: it places the metal at a precise coordination geometry that fixes  $\Phi$ , and it supplies the electron during catalysis.
3. **Dynamics** (femtosecond–picosecond side-chain and solvent fluctuations) stroke the active site, enabling the rapid H-like/He-like oscillation required by the cycle.

### 6.2 Case study: Cytochrome P450

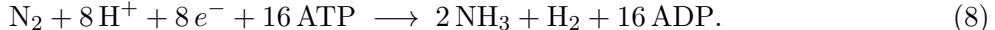
The reaction is oxidative hydroxylation:



The Fe–heme–thiolate center oscillates between  $\text{Fe}^{3+}$  ( $\Phi = 0.9$ ) and the oxoferryl intermediate **Compound I** ( $\text{Fe}^{4+}=\text{O}$ ,  $\Phi = 1.15$ ). Compound I is a near-optimal He-like acceptor: it abstracts the  $\chi_{\text{H}} = 4.8\text{eV}$  from the C–H bond, then rebounds the hydroxyl to the radical at  $\Phi \approx 0.9$  [4]. Point mutations that loosen or tighten the thiolate hydrogen bond shift  $\Phi$ ; mutations that tune it to exactly 1.0 increase turnover (Hypothesis H<sub>6</sub>, Section 9).

### 6.3 Case study: Nitrogenase

The reaction is nitrogen fixation under ambient conditions:



The FeMo-cofactor is a *distributed* Fluidity engine: Mo ( $\Phi \approx 1.2$ , relatively He-like) anchors and polarizes  $\text{N}_2$ , while the Fe atoms ( $\Phi \approx 0.8$ , relatively H-like) donate electrons in a conveyor belt that progressively breaks the  $\text{N}\equiv\text{N}$  triple bond. The cluster’s sulfur ligands buffer the oscillation, keeping the whole cofactor at  $\Phi \approx 1.0$  [5].

The theory makes a concrete structural prediction: substitution of Mo by V or Fe changes the *selectivity* (binding preference and product ratio) exactly because it changes the cofactor’s  $\Phi$  distribution, a shift that has been observed experimentally across the V– and Fe–nitrogenases.

### 6.4 Case study: Carbonic anhydrase

The biologically essential reaction is near diffusion-limited:



The active site is a  $\text{Zn}^{2+}$  coordinated by three histidines, resting at  $\Phi = 0.95$ . The metal polarizes the bound water (raising its  $\Phi$  from 1.6 toward 1.9), and the resulting Zn-bound hydroxide becomes strongly H-like ( $\chi_{\text{H}} = 2.5\text{eV}$ ), attacking  $\text{CO}_2$  ( $\chi_{\text{He}} = 6.0\text{eV}$ ) through the Zn-mediated bridge; the product bicarbonate ( $\Phi \approx 1.2$ ) is released [6]. Mutations that stiffen or soften the second-shell coordination raise Zn  $\Phi$  to 1.05 and increase turnover, again the signature of Fluidity tuning.

## 7 The Unified Catalytic Design Rule

The three domains collapse into one rule.

**Design Rule 1** (Unified Catalytic Design Rule). *An effective catalyst must:*

1. *maintain  $\Phi_{\text{cat}} \approx 1.0$  at the transition state; and*
2. *oscillate rapidly between  $\Phi \approx 0.8$  (H-like, electron-donating) and  $\Phi \approx 1.2$  (He-like, electron-accepting) throughout the cycle.*

*This rule applies identically to metal surfaces (Haber–Bosch), organometallic complexes (Grubbs), and enzymes (P450, nitrogenase, carbonic anhydrase).*

Consistent with the cross-domain map of the corrosion paper [10], the same chemical modulations recur: **fluorine doping poisons** catalysis (rigidifies the active site out of the CAW), **phosphate** acts as a promoter (moderates  $\Phi$ ), **chromium-alloying** raises surface  $\Phi$  (conferring both corrosion resistance and, at moderate loadings, catalytic activity), **PEG** acts as a protein-like scaffold for enzyme mimics (holding a metal at a fixed geometry, i.e., fixed  $\Phi$ ), and **metal substitution** ( $\text{Mo} \rightarrow \text{V} \rightarrow \text{Fe}$ ) is the universal Fluidity dial.

## 8 Computational Protocol

The catalyst screening workflow follows the core protocol [9], specialized to the active site:

1. **Compute**  $\chi_{\text{H}}$  of the intermediate bound to the catalyst: work function/VBM (surfaces) or HOMO (complexes), PBE+ $U$  where needed.
2. **Compute**  $\chi_{\text{He}}$  of the acceptor: electron-affinity/CBM by  $\Delta\text{SCF}$  [7].
3. **Compute**  $\Phi_{\text{cat}}$  of the active site in its ground and transition states: finite-field DFT (solids, cofactors) or QM/MM (enzymes), averaged over a thermal ensemble to obtain  $\sigma$ .
4. **Check the CAW:** if  $0.8 < \Phi < 1.2$ , the site is candidate-active; screen libraries by this filter first.
5. **Design:** tune  $\Phi$  toward 1.0 with ligands/promoters/mutations whose  $\Phi$  is known from the Periodic Table of Fluidity (K doped Fe, NHC ligands, thiolate weakening, Zn geometry).

Because the filter precedes any kinetics, it reduces the screening space by more than 99%: of the millions of plausible active-site configurations, only those with  $\Phi \in (0.8, 1.2)$  are worth computing further.

## 9 Testable Hypotheses

**Hypothesis 1** ( $H_5$ : The catalytic Goldilocks window). *Every catalyst family—surfaces, complexes, enzymes—shows maximum activity at  $\Phi_{\text{cat}} \approx 1.0$ , with activity dropping outside  $0.8 < \Phi < 1.2$ .*

Test: *Plot catalytic turnover vs. computed  $\Phi$  across a homologous metal series (or mutant series); the data should collapse onto the universal Gaussian of Eq. (3) with  $\sigma \approx 0.15$ .*

**Hypothesis 2** ( $H_6$ : Enzyme engineering by Fluidity). *Point mutations that move an enzyme active-site  $\Phi$  toward 1.0 (and not away) increase turnover by more than an order of magnitude.*

Test: *Mutate P450 or carbonic anhydrase variants designed to raise  $\Phi$  to 1.0; measure  $k_{\text{cat}}/K_M$  relative to wild type.*

## 10 Conclusion

TRT reframes catalysis as a single question about a single number. A catalyst is a Fluidity oscillator that must sit, at its moment of greatest power, at  $\Phi = 1.0$ —as ambivalent as a medium can be—and oscillate between the donor- and acceptor-sides through every cycle. The Gaussian rate law  $k_{\text{cat}} \propto \exp(-(\Phi - 1)^2/2\sigma^2)$  is the quantitative form of the Sabatier principle, and it applies without modification to ammonia synthesis on iron, metathesis on ruthenium, and hydroxylation in your liver. The enzyme is not a peculiar kind of catalyst; it is the entire field’s lesson, evolved: hold an active metal at  $\Phi \approx 1.0$ , stroke it with dynamics, and the reaction will flow. In TRT, industrial catalysis and microscopic life are the same craft—*Fluidity engineering*.

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